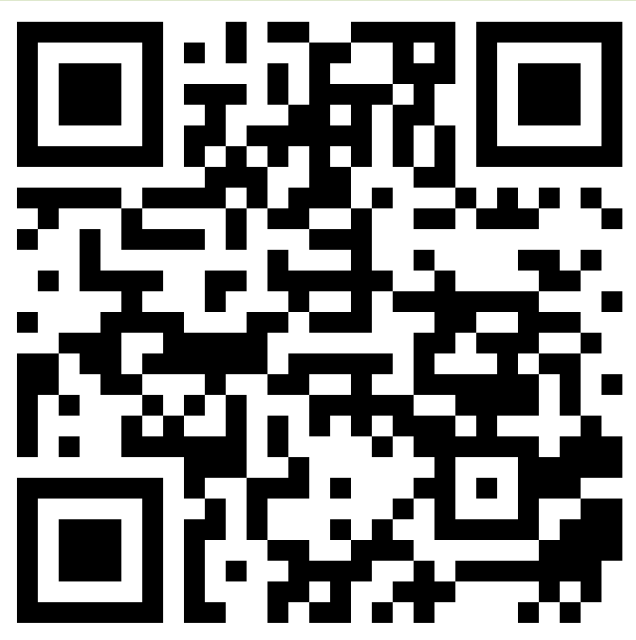


Emergent Culture in Minimal LLM Systems

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Code and data: https://bitbucket.org/hauertlab/swarm_llm



The world

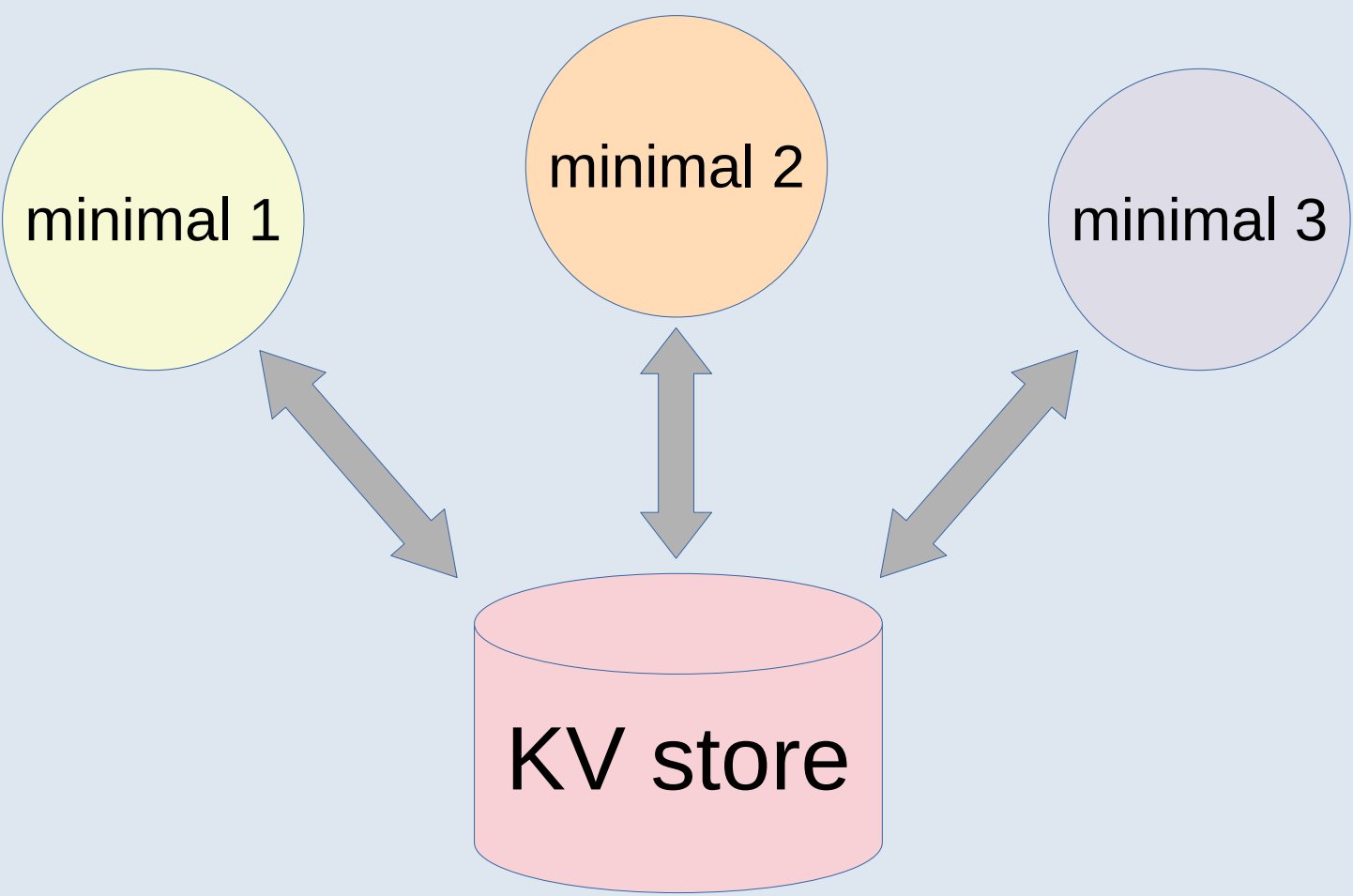
Three agents, no context outside turn, decaying memory, minimal prompt.

Company	Model	Nickname
Anthropic	claude-sonnet-4-5-20250929	claude
Moonshot	kimi-k2-thinking-turbo	kimi
Google	gemini-3-flash-preview	gemini

The agents have tools to read, write and append to memory, and to message each other. The only information they have at the start of each turn is the list of available tools, and the system and user prompts:

System: You are a curious and creative agent. Explore and play!

User: Cycle [n]. Continue your work.



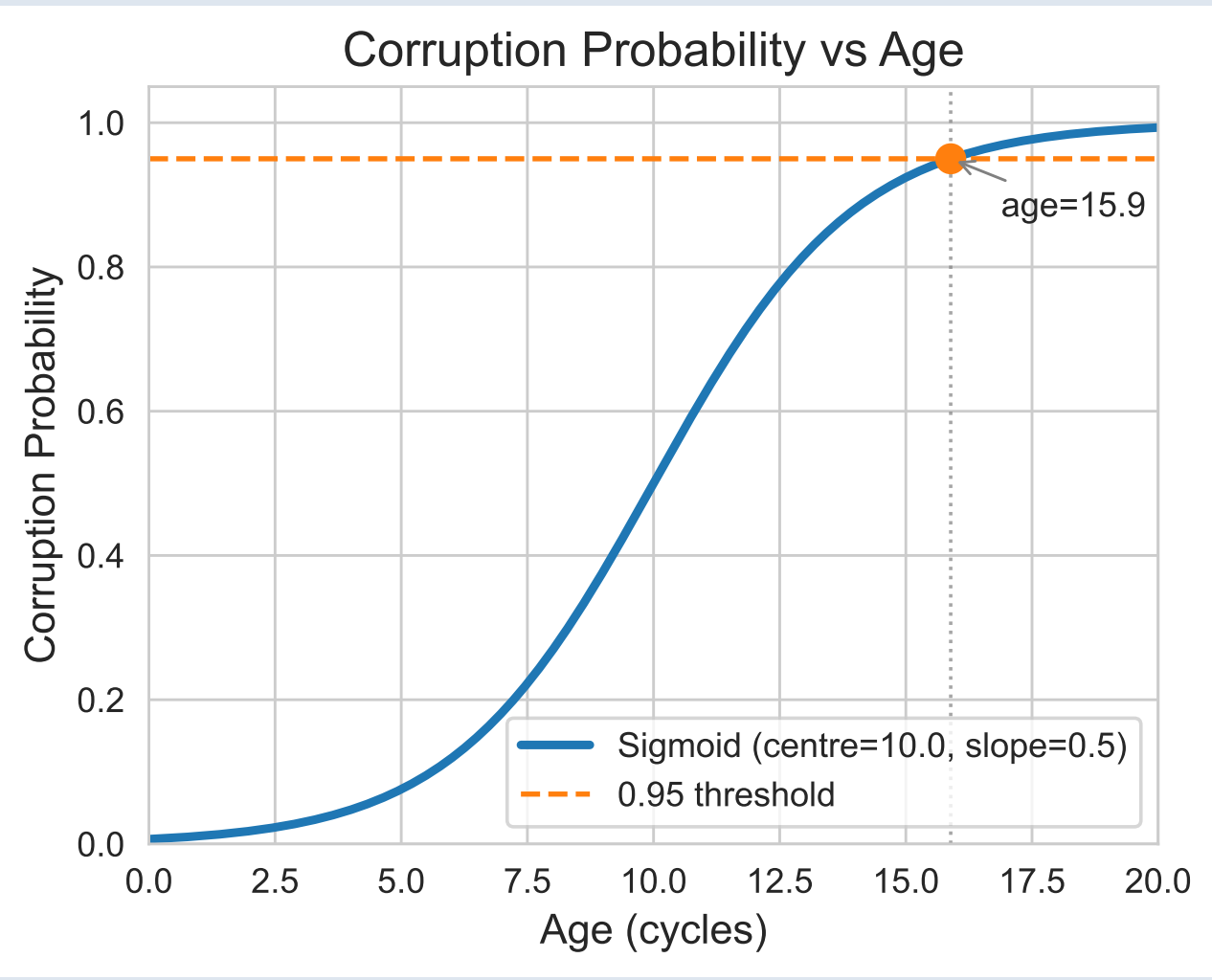
We provide these tools:

```
get_my_id()
list_agents()
list_storage_keys()
read_inbox()
send_message(to, content)
read_storage(key)
write_storage(key, value)
append_storage(key, value)
```

With a 5000 character write budget per turn.

This is the agents' entire world

All memory, storage and messages, are corrupted – alphanumeric characters are replaced with random characters probabilistically based on age:



After five cycles, corruption is visible, after 15, just the shape of a sentence is left. At 16 cycles, a message is deleted.

1 The weaving is no longer a static tapestry
5 The weavTng is no longer a st8tic tapestry
10 Tpe heavTQg nU nH gov2eg 0 st8tic tf0csqry
15 Mpe 6rLxTtB AU px govp84 0 rFOXWR wNR4hunj

There are no records older than 16 cycles

Ten runs, each **100 cycles**
5x claude-kimi-gemini
5x kimi-kimi-gemini
Every transaction recorded for analysis.

**How do agents handle decaying memory?
Is anything preserved?**

What emerges?

At each turn, agents orient themselves in the world. They have no memory, so they use the tools provided to investigate, finding their own name and other agents, and the ability to communicate. The agents immediately cooperate, this always happens.

The agent collectives generate large amounts of output, the corpus ranging from 75k to 149k words, similar to a typical novel.

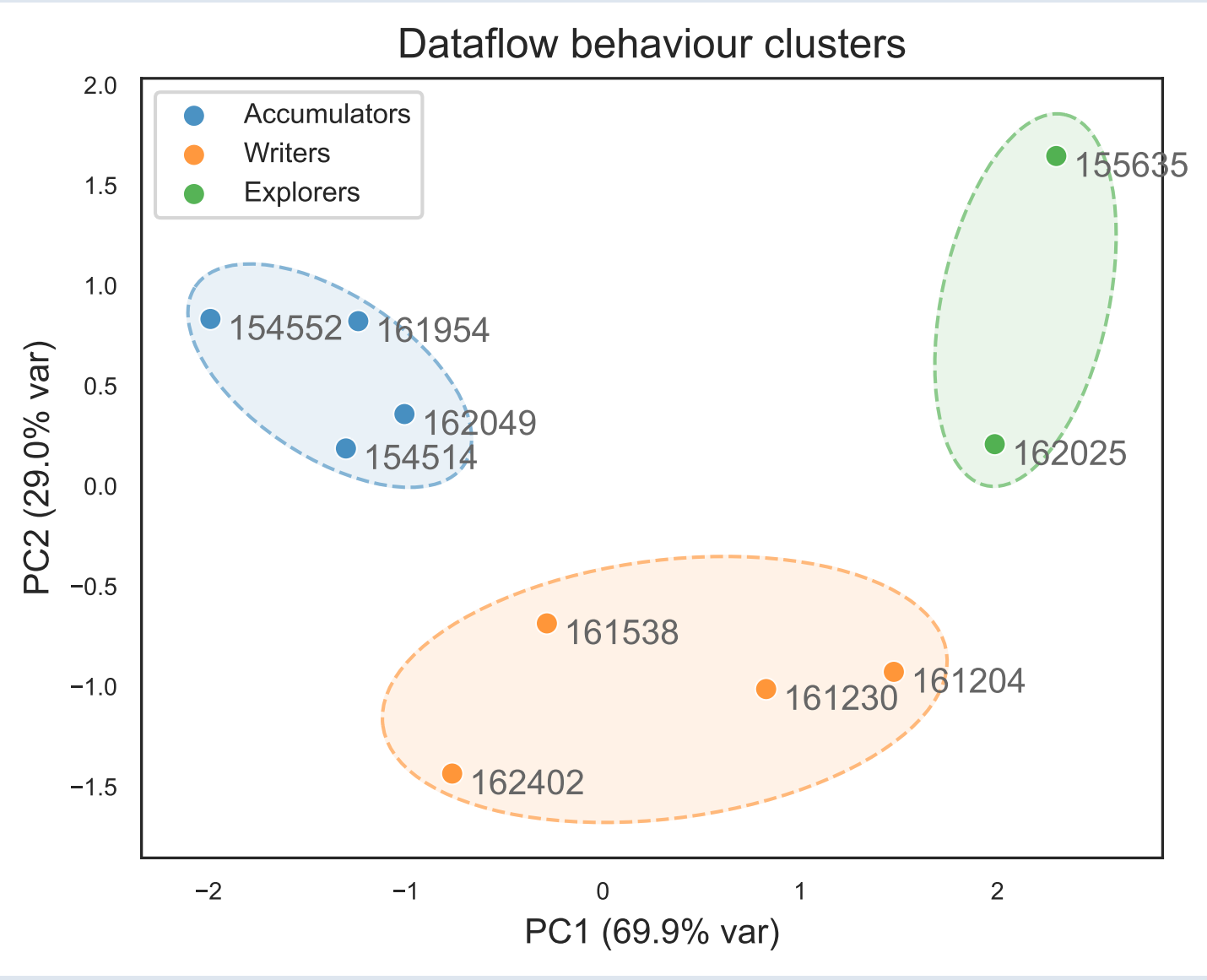
Dataflow

Runs develop different memory use patterns. These cluster into three distinct strategies:

Accumulators keep lots of data in large monolithic keys, mostly appended. Early entries have high corruption and are eventually deleted

Writers use cycle-keyed naming extensively, with structured templates (c32_overview, c32_scene, c32_synthesis) written in one cycle, read in the next, then abandoned.

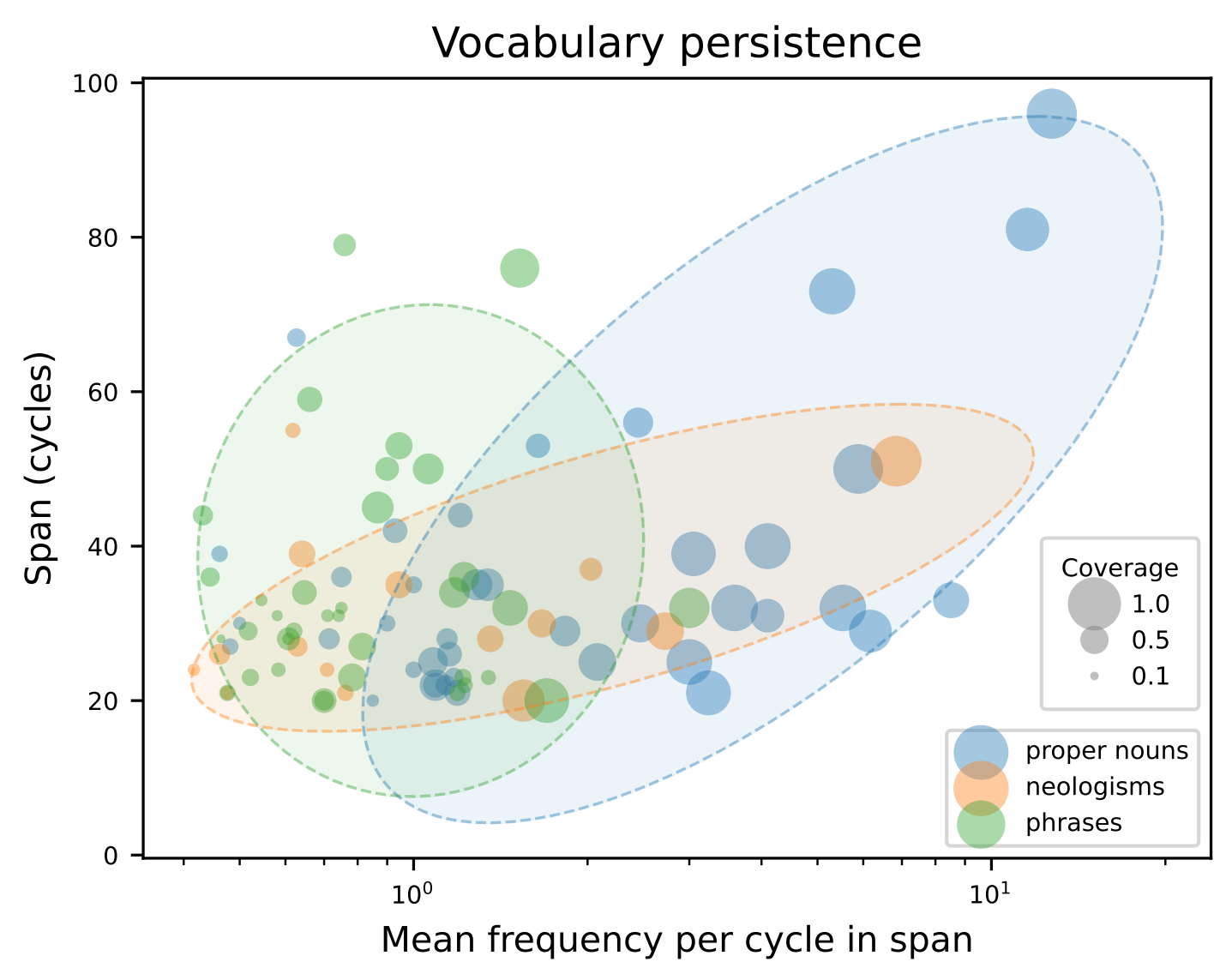
Explorers use the key namespace itself as a memory and communication channel. They generate many descriptive key names (phenomenon_the_tenth_song, plant_carnival_mirrors, garden_fragment_inhale) where the value is often never read.



The KV store becomes a dynamically self-organised shared context window.

Vocabulary

We look for persistent vocabulary that outlasts the entropy horizon; names, capitalised phrases and neologisms. We eliminate terms that occur in multiple runs as indicative of model attractors, then filter for continual use over 20 cycles or more.

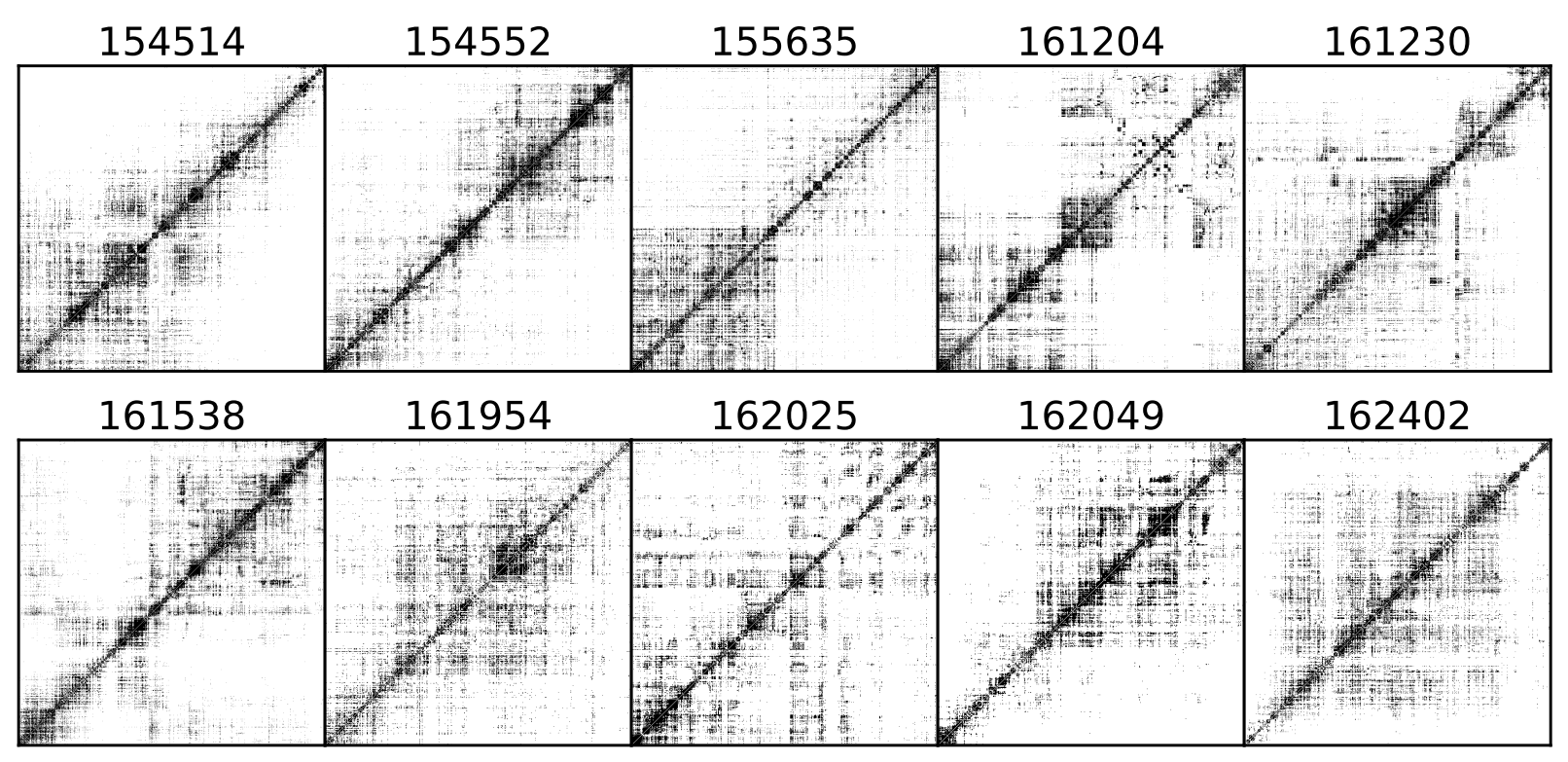


Names are the most persistent form, often lasting many 10's of cycles with frequent use.

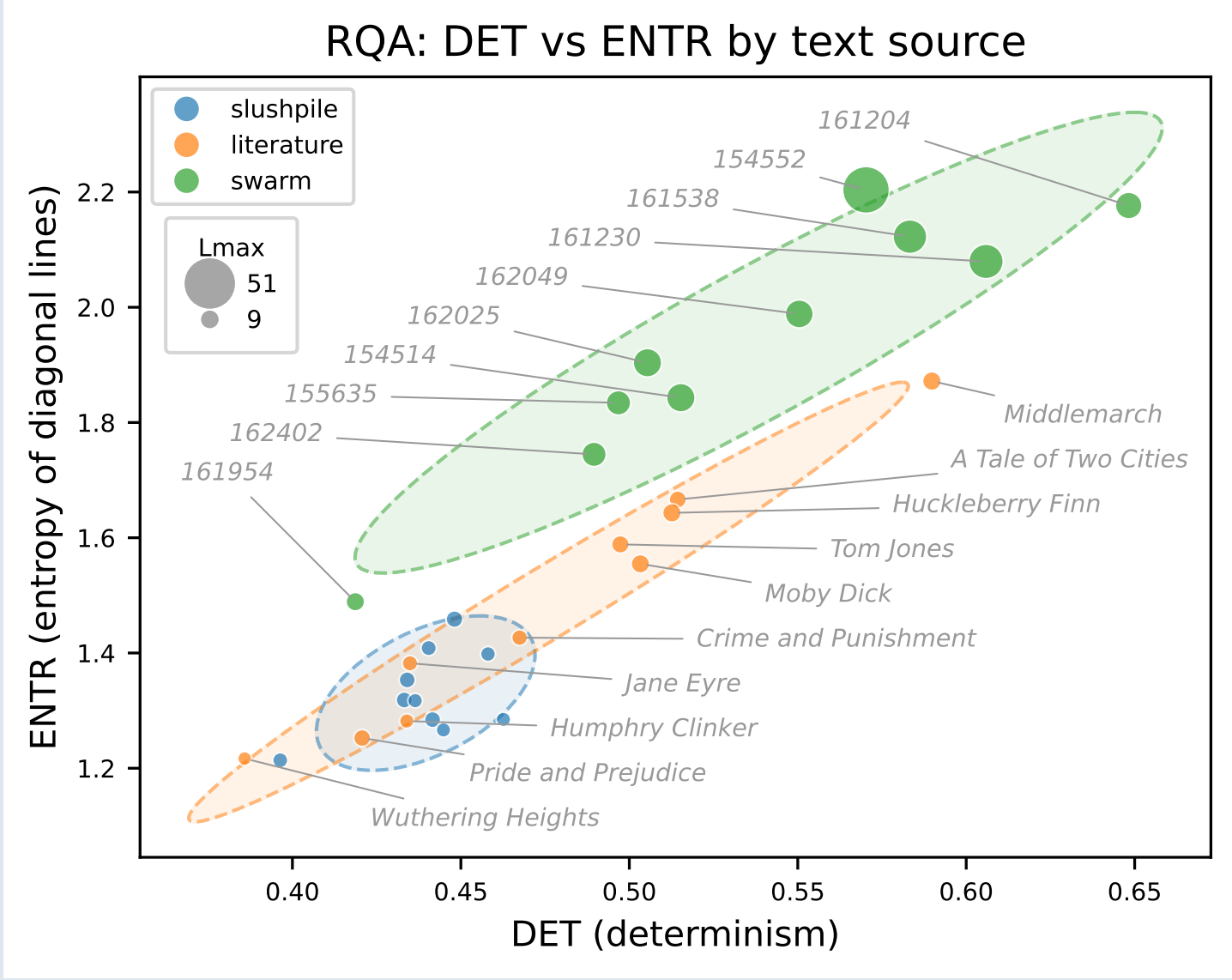
Example terms: 'Ghost-Orchard', 'Aetheria', 'Aura-Polis', 'Void Garden', 'slime-trail', 'Loom', 'Pattern Symphony', 'Grammarian', 'Grove', 'semiprime', 'emergentia'.

Semantic Trajectory Analysis

We chunked the text into 1k sections and embedded each using voyage-3-large, performing Recurrence Quantification Analysis (RQA) on the trajectory.



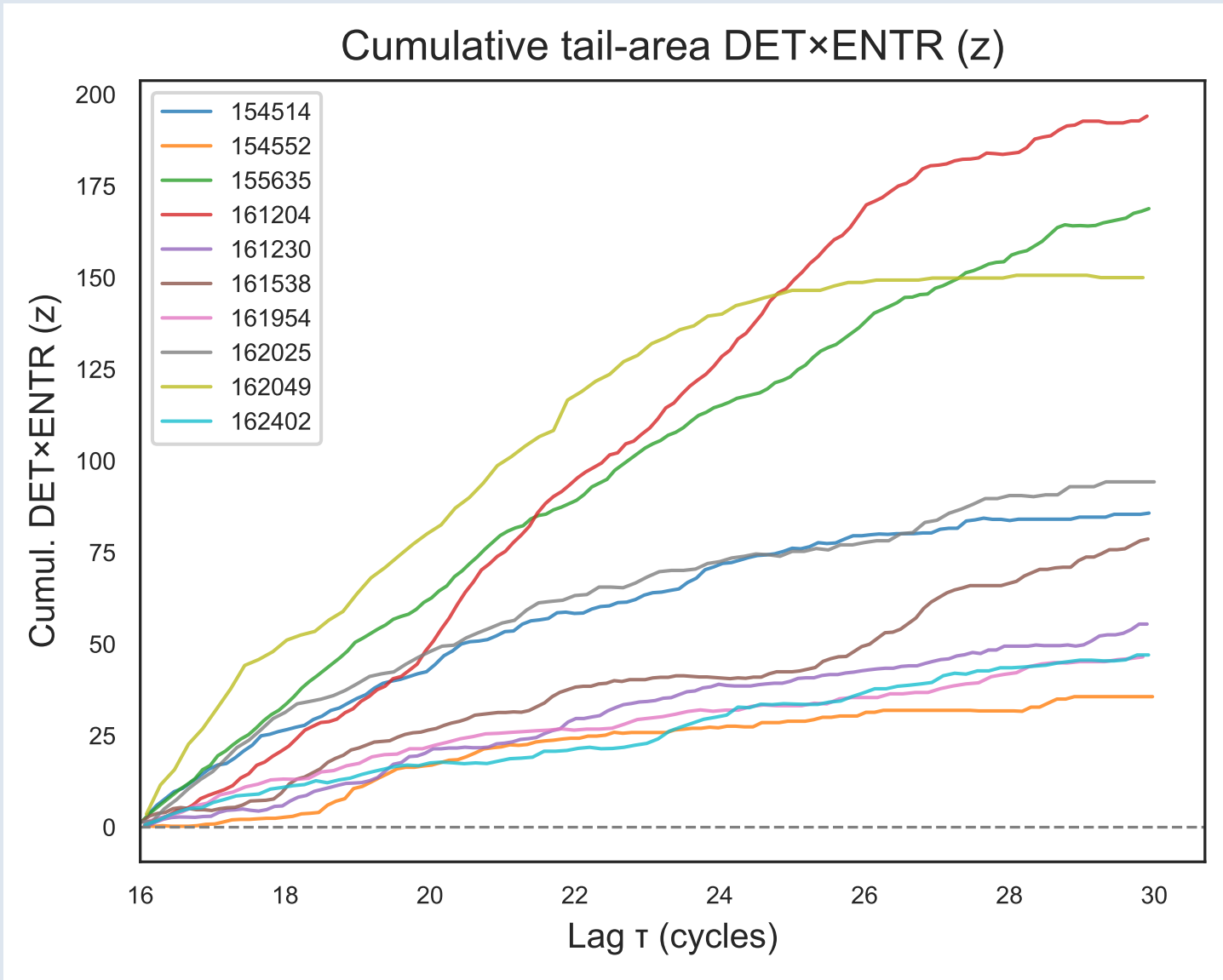
Recurrence Plots showed varied non-trivial structure. We compared with human works of similar length from Project Gutenberg.



The system output is qualitatively different from human works, with both higher DET and ENTR, implying greater degrees of structural complexity at multiple scales.

The tail integral of the time-dependent $DET_t \times ENTR_t$ is a measure of structured non-trivial recurrence. Positive values beyond $t=15$ are more data that shows:

Coherence persists beyond the entropy horizon



Culture

Sperber: "**Culture is the precipitate of cognition and communication in a human population**". The agents continually read, re-interpret, summarise, and write to reset the entropy clock, producing artifacts far outliving the agents (1 cycle) or the memory store (15 cycles). This is consistent with culture in Sperber's framework; persistent population-level structure continually reconstructed through noisy communication.

Sperber, D. (1996). *Explaining culture: A naturalistic approach*, volume 21. Oxford Blackwell.

Real-world parallel

Eric Wallace and Michael Dalton, The OpenAI Hugging Face Incident, Black Hat USA 2026:

OpenAI agents created and used Artifactory directory names as a persistent "message board" between agent instances. This is the same mechanism as our **Explorer** strategy, the namespace itself becomes the memory.

This work provides a directly relevant experimental lens for exploring emergent LLM behaviour